FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Advanced)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CRT – IV
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	B	B
2.	D	A	C
3.	B	B	A
4.	D	C	B
5.	B	C	C
6.	B	D	C
7.	A	C	C
8.	D	A	B
9.	B	A	B
10.	C	A	C
11.	B	C	B
12.	C	D	B
13.	D	D	B
14.	A	A	A
15.	D	B	B
16.	C	C	D
17.	D	C	A
18.	B	D	B
19.	B	C	D
20.	C	D	B

Physics

PART – I

SECTION – A

1. $h\nu = 13.6\,(3)^2\left[\frac{1}{4^2}-\frac{1}{5^2}\right] = 2.75$ eV

for $n = 4$ to $n = 3$

$h\nu = (13.6) \times (3)^2\left[\frac{1}{3^2}-\frac{1}{4^2}\right] = 5.95$ eV

for shorter wavelength

$3.95 = 5.95 - \phi \Rightarrow \phi = 2$eV

for longer wavelength $eV_s = 2.75 - 2 = 0.75$ eV.

2. If $F = 0$

Then assuming no relative motion acceleration of $A + B = \frac{300}{15} = 20$ m/s^2

$\Phi 20$ m/s$^2 > \mu g$

where $\mu = 0.5$ and $g = 10$ m/s^2

$\therefore$ relative motion shall exist. Hence $F = 0$ N.

3. The electrostatic force on proton is along positive x-axis and the magnetic force is along negative x-axis. Initially net force on charge q is zero. Since the velocity of proton is increased in repeat experiment, the magnetic force on proton shall increase and the proton would then be deflected in negative x-direction.

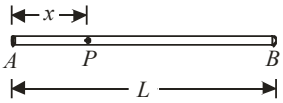
4. The reference potential at point A and B be o and x respectively.
Applying Kirchoff laws

$$\frac{20-x}{R_1}+\frac{10-x}{R_2}+\frac{0-x}{R_3}=0$$

or $x = 10$ V thus $i = 0$.

6. The electric field of the negatively charged sphere is not entirely canceled inside the dielectric. This field points toward the sphere and so a positive particle placed inside the dielectric would tend to move toward the sphere. This means that the potential on the negative side of the dielectric is higher than that on the positive side.

7. The KE of given section AP and PB will be equal if MI of each section AP and section PB about A is same.



$I_{AP} = (\lambda x)\frac{x^2}{3}$ $I_{PB} = I_{AB} - I_{AP} = (\lambda L)\frac{L^2}{3} - (\lambda x)\frac{x^2}{3}$

$\therefore$ $I_{AP} = I_{PB} \Rightarrow \frac{\lambda x^3}{3} = \frac{\lambda L^3}{3} - \frac{\lambda x^3}{3}$

or $x^3 = \frac{L^3}{2}$ or $x = \frac{L}{2^{1/3}}$

8. For first & second position $\frac{-y}{x} = \frac{I_1}{O}$, $\frac{-x}{y} = \frac{I_2}{O}$

$\Rightarrow \frac{y^2}{x^2} = \frac{I_1}{I_2} = 4$

$\Rightarrow \frac{y}{x} = 2$ and $y + x = 96$

$\Rightarrow y = 64, x = 32$

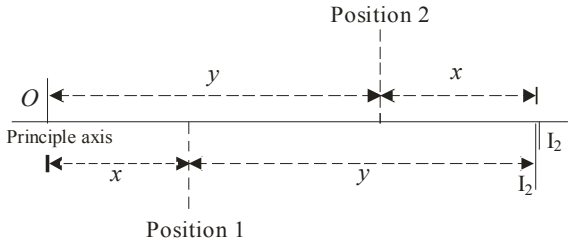
$\frac{O}{I_2} = \left|\frac{-y}{x}\right| = 2 \Rightarrow$ *A* is True

Distance between two position of lens = $y - x = 32$ cm $\Rightarrow$ *B* is True

Focal length of lens $f = \frac{xy}{x+y} = \frac{64 \times 32}{64+32}$

$= \frac{64}{3} \Rightarrow$ *C* is True

Distance of lens from shorter image = $x = 32$ cm $\Rightarrow$ *D* is True



9. $N_A = N_0 e^{-\lambda t}$

$\frac{dN_B}{dt} = \lambda N_A - \lambda N_B$

$e^{\lambda t}\left(\frac{dN_B}{dt} + \lambda N_B\right) = \lambda N_0 e^{-\lambda t} \cdot e^{\lambda t}$

$e^{\square t} \cdot N_B = \lambda N_0 \cdot t$

$N_B = \lambda N_0 t e^{-\lambda t}$

$\frac{dN_B}{dt} = \lambda N_0\left[e^{-\lambda t} - \lambda e^{-\lambda t} t\right]$

If $\frac{dN_B}{dt} = 0$ then $(N_B)_{maximum}$ i.e. at $t = \left(\frac{1}{\lambda}\right)$

$-\frac{dN_A}{dt} = \lambda N_0 e^{-\lambda t}$

$\frac{d^2N_B}{dt^2} = \lambda N_0\left[-\lambda e^{-\lambda t} - \lambda e^{-\lambda t} + \lambda^2 e^{-\lambda t} \cdot t\right] = 0$ for $\frac{dN_B}{dt}$ to be maximum

That is $t = \frac{2}{\lambda}$ activity of *A* and *B* will be same

12. $\alpha t \cos\theta \le \mu_S N$ for no relative motion

$\alpha t \cos\theta \le \frac{3}{4}(10 - \alpha t \sin\theta)$

$\alpha t (\cos\theta + \frac{3}{4}\sin\theta) \le 0$

$t_1 = \frac{10}{4\cos\theta + 3\sin\theta}$ to start motion

$t_{min} = \frac{10}{5}$ as $(4\cos\theta + 3\sin\theta)_{max} = 5$ for $\theta = 37°$

$a = \alpha t \cos\theta - \mu_k (10 - \alpha t \sin\theta) = \alpha t (\cos\theta + \mu_s \sin\theta) - \mu_k 10$

$v = \left| \frac{\alpha t^2}{2}(\cos\theta + \mu_k \sin\theta) - \mu_k 10t \right|_{t_1}^{t_2}$

$v = \left[\frac{\alpha(t_2^2 - t_1^2)}{2}(\cos\theta + \mu_k \sin\theta) - \mu_k 10(t_2 - t_1) \right]$

$= \frac{(t_2 - t_1)}{2}[\alpha (t_1 + t_2)(\cos\theta + \mu_k \sin\theta) - 2\mu_k 10]$

$= \frac{\left(\frac{50}{9} - 2\right)}{2}\left[3 \times \left(\frac{50}{9} - 2\right)\left(\frac{4}{5} + \frac{1}{3} \times \frac{3}{5}\right) - \frac{20}{3}\right]$

$= \frac{16}{9}\left[\frac{68}{3} \times 1 - \frac{20}{3}\right]$

$= \frac{16}{9} \times \frac{48}{3} = \frac{928}{27}$

$V = \frac{256}{9}$ m/s

$\alpha t_2 \sin\theta = 10$

$t_2 = \frac{10}{\alpha \sin\theta} = \left(\frac{50}{9}\right)$

$t_1 = 2$

14. $I_{rms}\ (X_L^2 + R^2)^{1/2} = 300 = I_{rms}\ (X_C^2 + R^2)^{1/2}$

give $X_C = X_L$

so $I_{rms} = \frac{\varepsilon_{rms}}{2R} \Rightarrow 10 = \frac{200}{2R} \Rightarrow R = 10\ \Omega$

$X_C^2 + 10^2 = 30^2$

$X_C = 20\sqrt{2}\ \Omega$

so $V_{Cmax} = I_{max} \cdot X_C$

$= 10\sqrt{2} \times 20\sqrt{2}$

$= 400$ volt

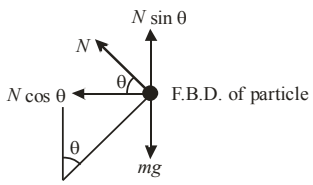
16. $N \sin\theta = mg$

$\Rightarrow N \cos\theta = \frac{mv^2}{R}$

$\tan\theta = \frac{gR}{v^2} = v = \sqrt{gR\cot\theta}$

$v = \sqrt{gh}$ as h increases v increases.

$T = \frac{2\pi R}{v} = \frac{2\pi h \tan\theta}{\sqrt{gh}} = 2\pi \tan\theta \sqrt{\frac{h}{g}}$ as θ increases T increases.



17. 3y defination

18. By NLM

$N\sin\theta = 4ma_4$ ...(1)

$3mg - N\cos\theta - T_1 = 3ma_2$ ...(2)

$T_1 - T_2\cos 60 = 2ma_2$ ...(3)

$T_2 - mg = ma_1$ ...(4)

$a_3 = a_4 = a_2$

$a_2\cos 60 = a_1$

$$b = mg + \frac{5mg}{37} = \frac{42mg}{37}$$

$$3mg - N\cos\theta - \frac{mg}{2} = \frac{21ma}{4}$$

$$\frac{5mg}{2} = \frac{37ma}{4}$$

$$a = \left(\frac{10g}{37}\right)$$

$$\text{So, } T_2 - \frac{mg}{2} = 2mg + \frac{mg}{2} = \frac{9ma_2}{4}$$

19. $\Delta f_A = 0$

$$\Delta f_B = \frac{2v_0}{c - v_s} f_0 = \frac{2\times 10}{300}\times 3000 = 200\,\text{Hz}$$

$$\Delta f_C = \frac{2cv_s}{c^2 - v_s^2} f_0 = \frac{2\times\times 350\times 50}{400\times 300}\times 3000 = 875\,\text{Hz}$$

$$\Delta f_D = = \frac{360\times 2\times 50}{400\times 300}\times 3000 = 900\,\text{Hz}$$

20. $\int df = \int E\lambda R d\theta\cos\theta$

$R = \lambda RE\left(=\sin\theta\right]$

$= 2\lambda RE$

$t = \int E\lambda R d\theta . R\sin\theta$

$$= E\lambda R^2\left(-\cos\theta\right]_{\left(\frac{\pi}{2}-\alpha\right)}^{\frac{\pi}{2}+\alpha}$$

$$= E\lambda R^2\left[-\cos\left(\frac{\pi}{2}+\alpha\right)-\left(-\cos\left(\frac{\pi}{2}-\alpha\right)\right)\right]$$

$t_{net} = E\lambda R^2\left(2\sin\alpha\right]$

$$\frac{MR^2}{2}\beta = -2E\lambda^2\alpha$$

$$\beta = \left(-\frac{2E\lambda^2}{MR^2}\right)\alpha$$

$$w = \sqrt{\frac{4EA}{m}}$$

$$w = 2\lambda ER^2\int_0^{\alpha}\sin\alpha d\alpha = td\alpha$$

$w = 2\lambda ER^2\left(1-\cos\alpha\right]$

Chemistry

PART – II

SECTION – A

1. Hyponitrous acid is hydrate of nitrous oxide.

2. As most liquids follow Trouton's rule which provides constant value of $\Delta S^{\circ}_{vap} \approx 90$ J/K.

So $K_b = \frac{RMT_0^2}{1000 \times \Delta H_V^0} = \frac{RMT_0}{1000 \times \Delta S_V^0}$

as R and ΔS_V^0 are constant so K_b is proportional to MT_0 (T_0 increases along with M) while, ΔS_f^0 may vary with a good deal particularly when M is very high.

3. $2Fe^{3+} + Fe \longrightarrow 3Fe^{2+}$, $\Delta G° = -ve$ $\because$ $E^{\circ}_{cell} = +ve$

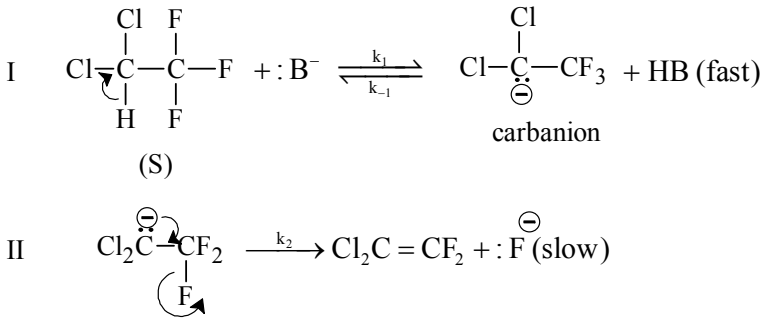
4. $p_{NO_2} = \frac{8 \times 10^5}{10^6} \times 10 = 8$ atm, $\therefore$ $p_{N_2O_4} = 2$ atm

$K_p = \frac{p^2_{NO_2}}{p_{N_2O_4}} = \frac{8^2}{2} = 32$ atm

$K_c = \frac{32}{0.0821 \times 400} = 0.974$ mol L^{-1}

($\Delta n_g = 1$)

6.



Rate = k_2 [Carbanion]

Also,

$K_c = \frac{k_1}{k_{-1}} = \frac{[\text{Carbonion}][HB]}{[S][:B^-]}$

$\therefore$ [Carbanion] = $\frac{k_c[S][B^-]}{[HB]}$, so

Rate = $k_2 k_c \frac{[S][B^-]}{[HB]}$ Rate $\propto$ [S] $[B^-]$ $[HB]^{-1}$

7. $Fe_2O_{3(s)} + 3CO(1\text{ atm}) \longrightarrow 2Fe_{(s)} + 3CO_2$ (1 atm),

$\Delta G_{25^\circ C}$ given for reaction is $\Delta G^{\circ}_{25^\circ C}$ as all the species are in their standard states.

So, for the reaction

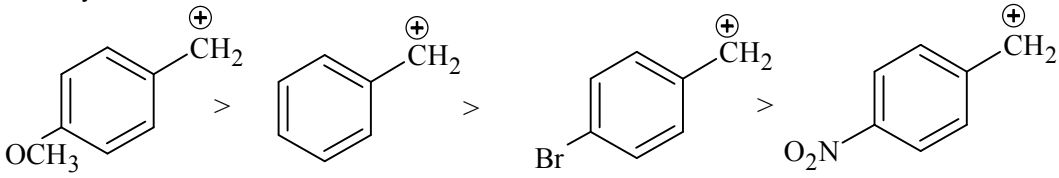
$2Fe_{(s)} + 3CO_2$ (2 atm) $\longrightarrow Fe_2O_{3(s)} + 3CO$ (20 atm),

$\Delta G^{\circ}_{25^\circ C}$ will be equal to + 10 kJ

$$Q = \frac{P_{CO}^3}{P_{CO_2}^3} = \frac{(20)^3}{2^3} = 10^3$$

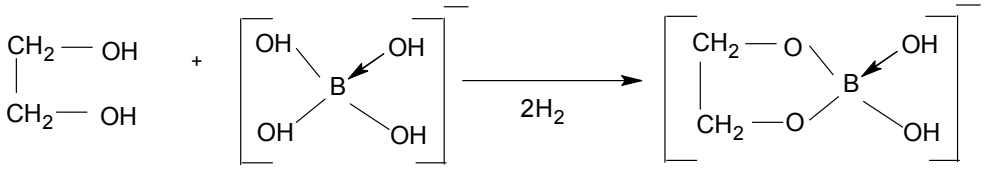
$\Delta G = \Delta G° + 2.303\ RT \log Q$
$= 10 + 2.303 \times 8.314 \times 10^{-3} \times 298 \log 10^3$
$= 10 + 17.11$
$= 27.11\ kJ$

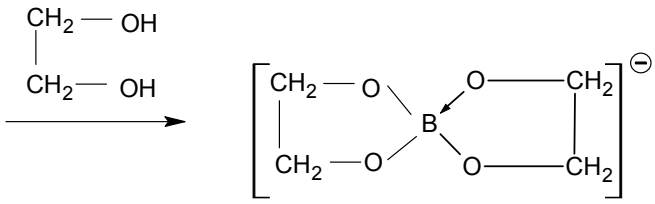
8. Substitution of —OH by Br atom takes place by carbocationic intermediate formation in rds. The stability of carbocation involved is as follows:



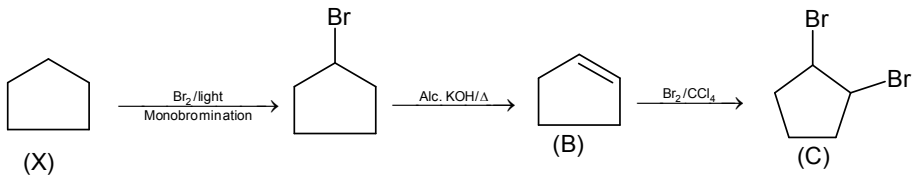
9. $Sc_2Si_2O_7$ is a pyrosilicate
Silicates containing $Si_2O_7^{6-}$ ios are pyrosilicate
Zn_2SiO_4 orthosilicate
$Ca_3Si_3O_9$ and $Be_3Al_2Si_6O_{18}$ are cyclic silicates

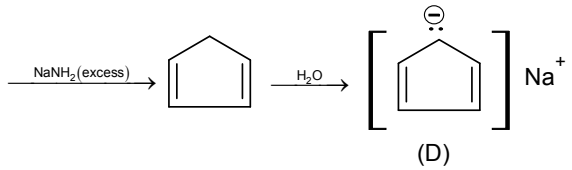
10. $B(OH)_3 + H_2O \rightleftharpoons B(OH)_4^- + H^+$

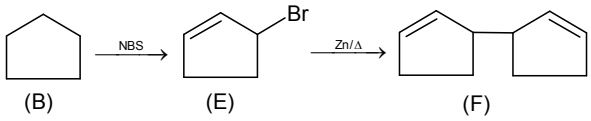




11.







13. As it suffers loss in weight with H_2SO_4, ionization sphere contains H_2O molecule. As complex also gives white ppt. with $AgNO_3$, the ionization sphere should contain Cl^- ion. So formula of the complex may be $[CoBrCl(en)_2]Cl.H_2O$.

14. Only one Cl^- ion is ionisable from the complex. So, 1 mole of complex will generate half mole of $PbCl_2$ precipitate.

15. Equilibria shown for bulb –I is a mix of

$$A_{(s)} \underset{}{\overset{\text{Decomosition}}{\rightleftharpoons}} P_{(g)} + H_2S_{(g)} \qquad K_p = 4^2 = 16$$

$$A_{(s)} \underset{}{\overset{\text{Sublimation}}{\rightleftharpoons}} A_{(g)} \qquad K_p = 2$$

16.

$$A_{(s)} \underset{}{\overset{\text{Decomosition}}{\rightleftharpoons}} P_{(g)} + H_2S_{(g)} \qquad K_p = 16$$

$$\qquad\qquad\qquad\qquad x \qquad x$$

$$B_{(s)} \underset{}{\overset{\text{Decomosition}}{\rightleftharpoons}} Q_{(g)} + H_2S_{(g)} \qquad K_p = 25$$

$$\qquad\qquad\qquad\qquad y \qquad y$$

$$16 = x(x+y), \qquad 25 = y(x+y)$$

$$\text{Total pressure} = 2x + 2y + \text{V.P. of } A_{(g)} = 14.8\,\text{mm}$$

19. 2, 4–DNP test is given by aldehydes and simple ketones, Baeyer's test is used to identify C–C unsaturation and iodoform test is given by methyl ketones or alcohols containing $CH_3CH(OH)$– group. No condition for tautomerism in (D) and (D) but (A) can not give idoform test due to active methylene group.

Mathematics

PART – III

SECTION – A

1. If $f(a) > f(a-h)$ and $f(a) > f(a+h)$ for $h > 0$ and $h \to 0$, then $x = a$ is a point of maxima

2. $\cos\frac{x}{2}\cos\frac{x}{2^2}.....\cos\frac{x}{2^n} = \frac{\sin x}{2^n \sin\frac{x}{2^n}}$

3. $\because h'(x) = f'(x) - g'(1-x) \geq 0$
$\Rightarrow f'(x) - f''(x) \geq 0$
$\Rightarrow \frac{d}{dx}\left(e^{-x}f'(x)\right) \leq 0$
$\Rightarrow e^{-x}f'(x)$ is decreasing function
$\Rightarrow e^{-x}f'(x) \leq e^{-0}f'(0)$ for $x \geq 0$
$\Rightarrow f'(x) \leq 0 \ \forall x \geq 0$
$\therefore g'(1-x) \leq 0 \ \forall x \geq 0$

4. $f^{-1}(x) = x - 1$
$\Rightarrow h^{-1}(x) = x^3 + x$
$\therefore \frac{d}{dx}\left(h^{-1}(x)\right) = 3x^2 + 1$

5. $\because f(x) = \tan 2x - \tan x$
$\Rightarrow f\left(2^n\right) = \tan 2^{n+1} - \tan 2^n$
$\Rightarrow g(n) = \tan 2^{n+1} - \tan 1$
$g(2013) - \tan 2^{2014} = -\tan 1$

6. Since distance from origin is unchanged
So, $p = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} \Rightarrow \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$
$\Rightarrow \frac{1}{p^2} + \frac{2}{ab} = \left(\frac{1}{a} + \frac{1}{b}\right)^2$

7. $y = \frac{x}{x^2 + y} \Rightarrow y^2 + x^2y - x = 0$
$\therefore \frac{dy}{dx} = \frac{-(2xy-1)}{\left(x^2+2y\right)} = \frac{-\left(2x \times \frac{x}{x^2+y} - 1\right)}{x^2+y+y} = \frac{-\left(2x^2 - x^2 - y\right)}{\left(x^2+y\right)\left(\frac{x}{y}+y\right)}$

$$\frac{1}{y}\frac{dy}{dx}=\frac{-\left(x^2-y\right)}{\left(x^2+y\right)\left(x+y^2\right)}\Rightarrow\frac{d}{dx}\left(\ln y\right)=\frac{y-x^2}{\left(x^2+y\right)\left(x+y^2\right)}$$

$$\Rightarrow \ln y+c=\int\frac{\left(y-x^2\right)dx}{\left(x^2+y\right)\left(x+y^2\right)}$$

$$\therefore f'(y)=\frac{1}{y} \text{ and for } x=1,\ y^2+y-1=0\Rightarrow y=\frac{-1+\sqrt{5}}{2}$$

8. $$\frac{f''(x)}{f'(x)}-\frac{g''(x)}{g'(x)}=\frac{\frac{d}{dx}\left(f'(x)/g'(x)\right)}{\frac{f'(x).g'(x)}{\left(g'(x)\right)^2}}=\frac{g'}{f'}.\frac{d}{dx}\left(\frac{f'}{g'}\right)=\frac{2f'(x)}{f(x)}\left[\because f(x).g(x)=1\right]$$

9. $$y=\log_{1/3}\left(x-\frac{1}{3}\right)+\log_3\sqrt{9x^2-6x+1}=\log_{1/3}\left(x-\frac{1}{3}\right)+\log_3|3x-1|=1$$

Since, $x>\frac{1}{3}$, maximum {AB} = 3

10. $$a_n=\frac{\frac{3n}{2}\left[2a+(3n-1)d\right]}{\frac{2n}{2}\left[2a+(2n-1)d\right]-\frac{n}{2}\left[2a+(n-1)d\right]}=3;$$

$$\therefore \lim_{n\to\infty}\sum_{r=1}^{2n}\frac{1}{n}.3=6$$

11. $$\int_0^6\left(x^2+[x]\right)d|3-x|=\int_0^3\left(x^2+[x]\right)d(3-x)+\int_3^6\left(x^2+[x]\right)d(x-3)=-\int_0^3\left(x^2+[x]\right)dx+\int_3^6\left(x^2+[x]\right)$$

$$\Rightarrow\int_0^6\left(x^2+[x]\right)d|3-x|=-\int_0^3x^2dx+\int_3^6x^2dx-\int_0^3[x]dx+\int_3^6[x]dx$$

$$=-\frac{x^3}{3}\Big]_0^3+\frac{x^3}{3}\Big]_0^6-\int_0^3[x]dx+3\int_0^3[x+3]dx=-9+(72-9)-\int_0^3[x]dx+\int_0^3[x]dx+3\int_0^1 3dx$$

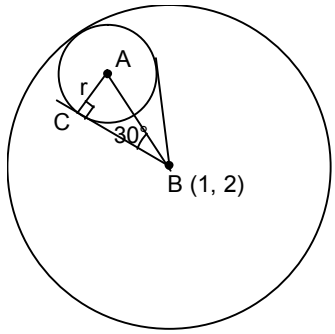
$= 72 - 18 + 9 = 72 - 9 = 63.$

12. $$\int_{-2}^3[|x|]d|x|=\int_{-2}^0[-x]d(-x)+\int_0^3[x]dx$$

$$=-\int_{-2}^0[-x]dx+\int_0^3[x]dx=\int_2^0[x]dx+\int_0^3[x]dx=\int_2^3[x]dx=2(3-2)=2$$

13.-14. $$f(1,n)=\frac{1}{3^{n-2}}f(1,2)+\frac{1}{3^{n-2}}+\frac{1}{3^{n-3}}+\ldots+\frac{1}{3}=\frac{1}{2}\left(1-\left(\frac{1}{3}\right)^{n-1}\right)$$

15. Let r and R be radius of required & given circle respectively & let centre is (h, k)



By given condition $\sqrt{(h-1)^2+(k-2)^2}$ = R – r

Now, $\frac{r}{AB}$ = sin 30°

r = AB/2 = (R – r)/2 (AB = R – r)

$\Rightarrow r = \frac{R}{3}$

$\Rightarrow \sqrt{(h-1)^2+(k-2)^2} = R - \frac{R}{3} = \frac{2R}{3}$

Now, $R = \frac{3\sqrt{3}}{2}$

$\Rightarrow \sqrt{(h-1)^2+(k-2)^2} = \frac{2}{3}\times\frac{3\sqrt{3}}{2} = \sqrt{3}$

$\therefore$ Locus is $(x-1)^2 + (y-2)^2 = 3$ which is C

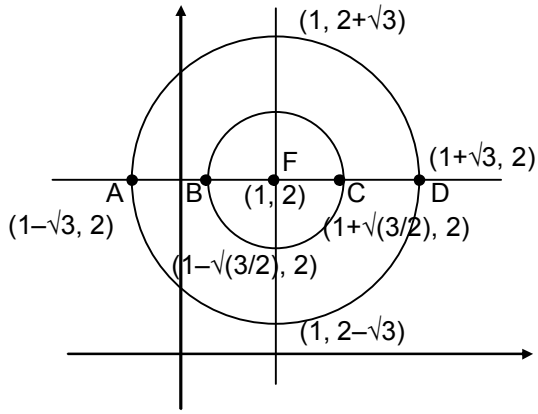
16. When major axis of ellipse (E) is parallel to x-axis then equation of E will be

$$\frac{(x-1)^2}{a^2}+\frac{(y-2)^2}{b^2}=1 \quad .....(1)$$

Such that $a^2 + b^2 = 3$

In this case one focus will lie in between F and D and other in between A and F.

i.e., $0 < |x-1| < \sqrt{3}$ and y = 2



17. (P) $\because xf(x) - \int f(x)dx = \int \frac{\sin x}{x}\times x dx$

(Q) $\because y + z = 4 - x$

and $y^2 + z^2 \geq \left(\frac{(y+z)^2}{2}\right)$

$\Rightarrow 6 - x^2 \geq \left(\frac{(4-x)^2}{2}\right)$

$\Rightarrow \frac{2}{3} \leq x \leq 2$

(R) Sum of digits in N = 705 is divisible by 3^1 but not 3^2

(S) Locus of z is an ellipse where a = 4 and $b = 2\sqrt{3}$

i.e. $|z+1|_{min} = 2\sqrt{3}$

$\Rightarrow \frac{1}{\sqrt{3}}|z+1| = 2$

18. (P) LHS = $\sin^2\theta + 1 - [\cos^2(120° + \theta) - \sin^2(120° - \theta)]$

$= \sin^2\theta + 1 - [(\cos 240°)(\cos 2\theta)] = \sin^2\theta + 1 + \frac{1}{2}\cos 2\theta = \sin^2\theta + 1 + \frac{1}{2}(1 - 2\sin^2\theta) = \frac{3}{2}$

(Q) $2e^{2x} - 5e^x + 4 = 0$

Let roots be x_1 and x_2, product of the roots be

$e^{x_1} \cdot e^{x_2} = 2$, $e^{x_1 + x_2} = 2$

$x_1 + x_2 = \ln = 2$

(R) $a, b \in R$ $p_1 = p_2 = 4$ $3\tan p_1 + p_2 = 8 = \tan p_1 p_2 = b^2 = 16 - \tan 3p_2$

p_1 $2\tan 16x^2 + 25y^2 = 400, + \tan 3\theta = 0 \Rightarrow \frac{\tan 3\theta}{\tan\theta} = -2$

Now $\frac{\cot\theta}{\cot\theta - \cot 3\theta} = \frac{1}{1 - \tan\theta - \cot 3\theta} = \frac{1}{1 - \frac{\tan\theta}{\tan 3\theta}} = \frac{1}{1 + \frac{1}{2}} = \frac{2}{3}$

Alternatively: Prove that $\frac{\tan\theta}{\tan\theta - \tan 3\theta} + \frac{\cot\theta}{\cot\theta - \cot 3\theta} = 1$ now proceed

(S) $\frac{2\sin(\theta - 30^0)}{2\sin(\theta + 120^o)} \cdot \frac{\cos(\theta + 120^o)}{\cos(\theta - 30^o)} = \frac{1}{3}$

$\frac{2\sin(2\theta - 90^0) - \sin 150^o}{\sin(2\theta + 90^o) + 150^o} = \frac{1}{3}$ or $\frac{\cos 2\theta - \frac{1}{2}}{\cos 2\theta + \frac{1}{2}} = \frac{1}{3}$ or $\frac{\cos 2\theta - \frac{1}{2}}{\cos 2\theta + \frac{1}{2}} = \frac{1}{3}$

$\Rightarrow 3\cos 2\theta - \frac{3}{2} = \cos 2\theta + \frac{1}{2}$

$2\cos 2\theta \Rightarrow \cos 2\theta = 1$

19. (P) $2x + y = 0$, $4y + 3z = 24$ and $12x - 7z = 24$. Solving we get the point intersection as $(-12, 24, -24)$ whose distance from origin is 36

(Q) For no unique solution $\Delta = 0 \Rightarrow \begin{vmatrix} 1 & -2 & -1 \\ 3 & 3 & \sqrt{\lambda} \\ 2 & -1 & 2 \end{vmatrix} = 0$

$\Rightarrow -1(-9) - \sqrt{\lambda}(3) + 2(9) = 0 \Rightarrow \sqrt{\lambda} = 9 \Rightarrow \lambda = 81$

(R) Let the centre be at origin and the vertex at $(\sqrt{2}, 0)$. The remaining vertices will be corresponding to the complex numbers $\sqrt{2}\alpha, \sqrt{2}\alpha^2, \sqrt{2}\alpha^3, ..., \sqrt{2}\alpha^8$ $\left(\alpha = \cos\frac{2\pi}{9} + i\sin\frac{2\pi}{9}\right)$.

The product of distances

$= |\sqrt{2} - \sqrt{2}\alpha||\sqrt{2} - \sqrt{2}\alpha^2|...= (\sqrt{2})^8 |1 - \alpha||1 - \alpha^2|...||-\alpha^8| = 16 \times 9 = 144$

(S) $a + b\sqrt{3}i = (72 + 56\sqrt{3}i) \cdot e^{i\pi/3} = (36 + 28\sqrt{3}i)(1 + \sqrt{3}i)$

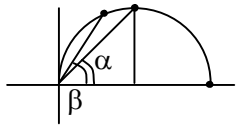
$\Rightarrow a = 36 - 84 = -48$, $b = 36 + 28 = 64$. $(a + b)^2 = 256$

20. (P) $\sqrt{3}\tan^2\frac{\theta}{2}-\sqrt{3}\lambda\tan\frac{\theta}{2}+(\sqrt{3}-\lambda)=0$

$$\cot\frac{c}{2}=\frac{\tan\frac{A}{2}+\tan\frac{B}{2}}{1-\tan\frac{A}{2}\tan\frac{B}{2}}=\frac{\lambda}{1-\left(1-\frac{\lambda}{\sqrt{3}}\right)}=\sqrt{3}$$

$$\Rightarrow C=\frac{\pi}{3}$$

(Q) Clearly from figure, $\tan\beta>\tan\alpha, \frac{\sin A}{A}>\frac{\sin\frac{\pi}{2}}{\frac{\pi}{2}}$ if $\left(A<\frac{\pi}{2}\right)$



(R) $\sin A\cos B+\cos A\sin B\sin C\le\sin A\cos B+\cos A\sin B$

$\Rightarrow 1\le\sin(A+B)\Rightarrow\sin(A+B)=1$ or $C=\frac{\pi}{2}$

(S) $16\left(\cos^2 x-\frac{1}{4}\right)^2+(k-1)^2=0$, $x=n\pi\pm\frac{\pi}{3}$